Terraform CI/CD Pipeline for AWS Infrastructure Automation

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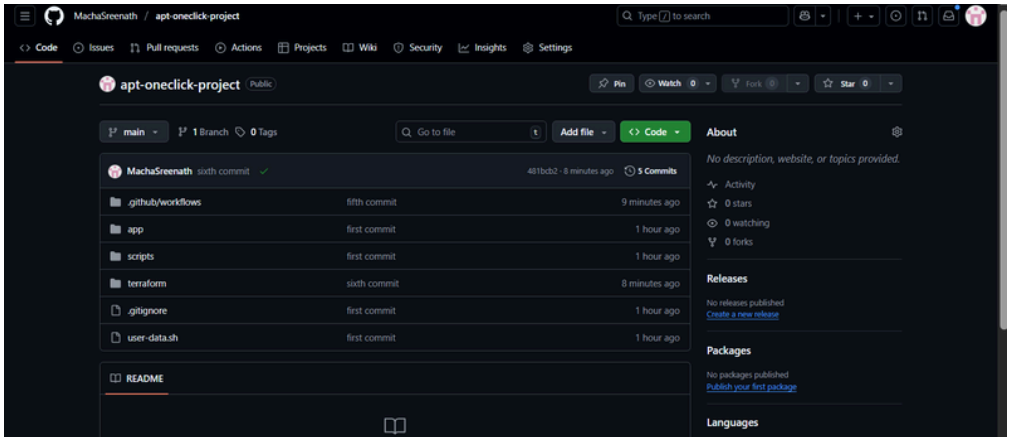
Architecture Overview:

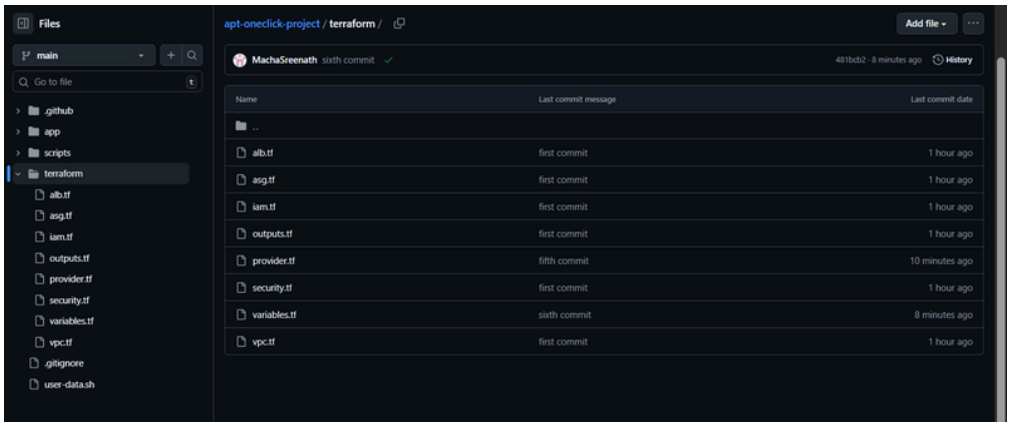
This project demonstrates a fully automated CI/CD pipeline using GitHub Actions and Terraform. It provisions AWS resources including VPC, Subnets, EC2 instances, Auto Scaling Groups, Security Groups, and an Application Load Balancer. The workflow allows manual deployment (apply) or deletion (destroy) using GitHub Actions.

```
Client → ALB (public subnets) → Target Group → ASG → EC2 (private subnets)
                                        |
                                NAT Gateway (egress)
                                        |
                                Internet Gateway
```

Repo Structure

Repository Structure and Important Files





Terraform Code Overview

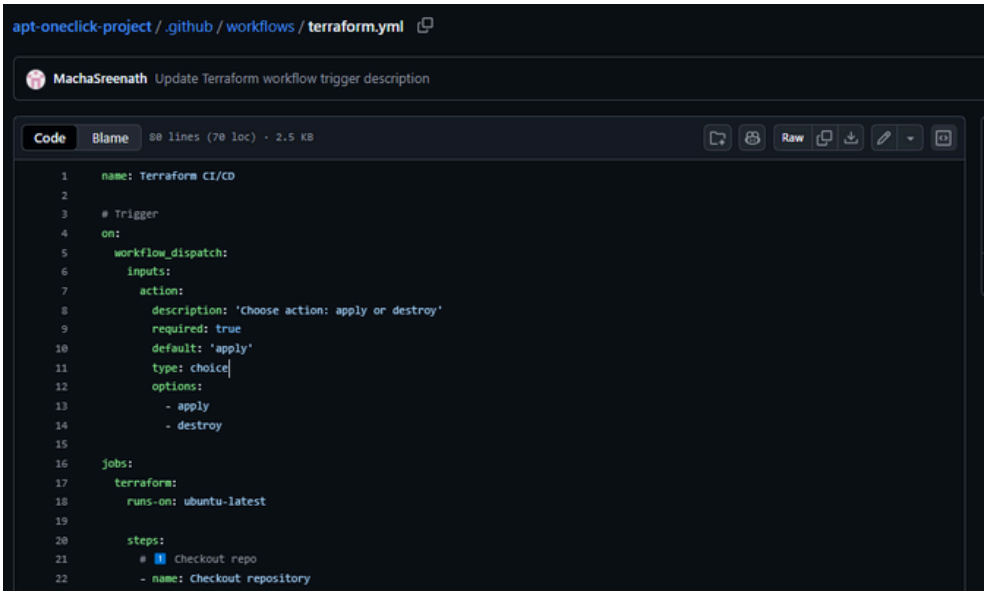
Key Terraform Files and Their Purpose

provider.tf	Configures AWS provider and region. Uses GitHub Secrets for credentials.
variables.tf	Declares variables like AWS region, VPC CIDR, instance type, and ASG settings.
vpc.tf	Creates the VPC and subnets.
security.tf	Defines Security Groups for EC2 and ALB.
alb.tf	Sets up Application Load Balancer and Target Groups.
asg.tf	Configures Auto Scaling Group and EC2 instances.
iam.tf	Creates IAM roles and policies if required.
outputs.tf	Defines Terraform outputs like ALB DNS name, VPC ID.

GitHub Actions Workflow

CI/CD Workflow Overview

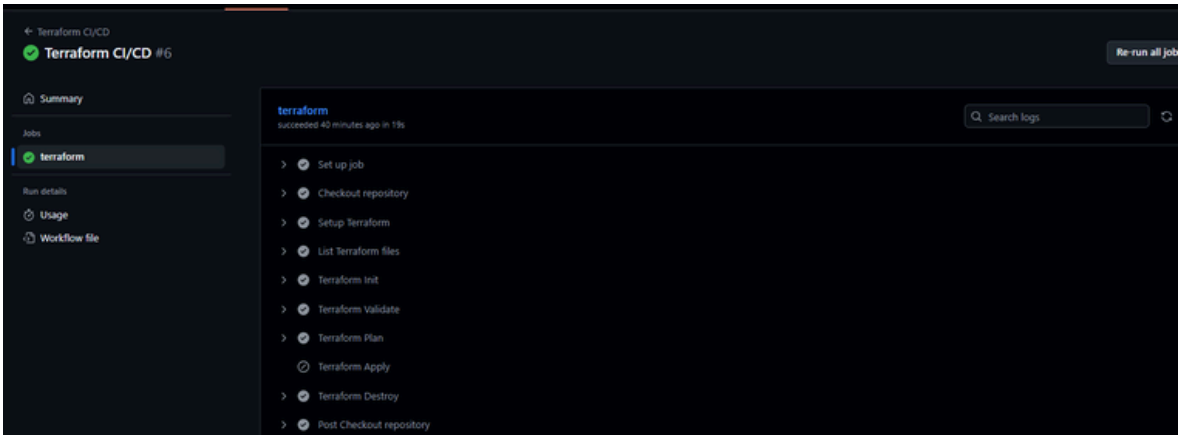
Checkout repository → Pulls the code from GitHub.
Setup Terraform → Installs Terraform CLI.
Init & Validate → Initializes Terraform working directory and validates configuration.
Plan → Shows what changes Terraform will apply.
Apply / Destroy → Manual workflow dispatch: user chooses action to deploy or delete resources.

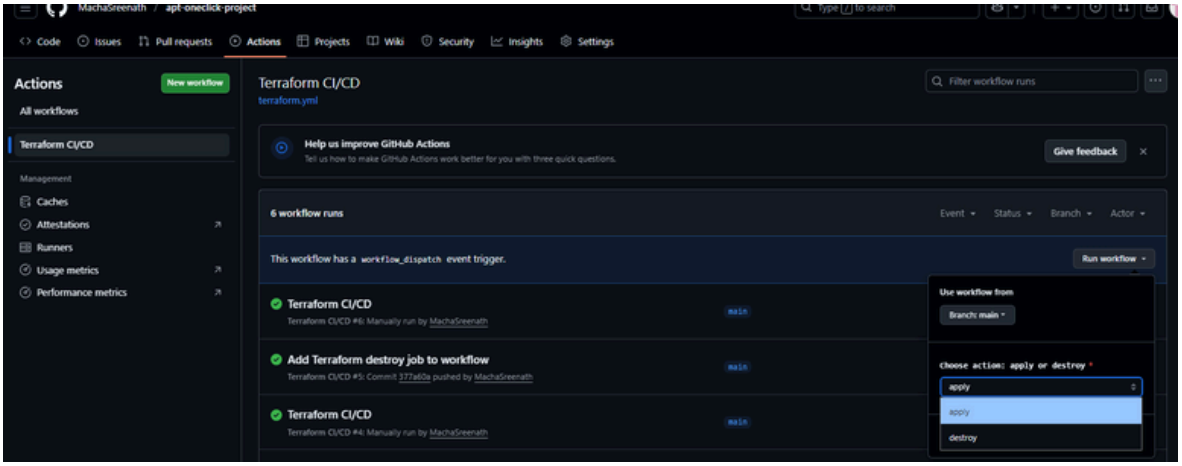


Workflow Run / Execution

Terraform Workflow Execution

Go to Actions → Terraform CI/CD → Run workflow
Select branch main
Choose action: apply or destroy





AWS Resources Verification

Verify Infrastructure in AWS Console

VPC → AWS VPC dashboard
Subnets → AWS VPC → Subnets
EC2 Instances → Running instances
Auto Scaling Group → EC2 → Auto Scaling
ALB / Target Group → EC2 → Load Balancers

Your VPCs

VPCs | VPC encryption controls - *new*

Your VPCs (2) Info

Last updated less than a minute ago | Actions | Create VPC

Find VPCs by attribute or tag

Name	VPC ID	State	Block Public Acc...	IPv4 CIDR	IPv6 CIDR	DHCP option s
apt-devops-vpc	vpc-029fdec5587e5bf67	Available	Off	10.0.0.0/16	–	dopt-07e6d78
–	vpc-0979ce9ab31158a9f	Available	Off	172.31.0.0/16	–	dopt-07e6d78

Subnets (10) Info

Last updated less than a minute ago | Actions | Create subnet

Find subnets by attribute or tag

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv
–	subnet-0ccf64b4b868e8797	Available	vpc-0979ce9ab31158a9f	Off	172.31.0.0/20	–
–	subnet-0a0a06c52cc1352c2	Available	vpc-0979ce9ab31158a9f	Off	172.31.80.0/20	–
–	subnet-0703ebd38f263a6cf	Available	vpc-0979ce9ab31158a9f	Off	172.31.32.0/20	–
–	subnet-00d491ec671dfe947	Available	vpc-0979ce9ab31158a9f	Off	172.31.48.0/20	–
public-1	subnet-0d572efb03466425e	Available	vpc-029fdec5587e5bf67 | apt-devops-...	Off	10.0.1.0/24	–
private-0	subnet-0f35d70391102ba62	Available	vpc-029fdec5587e5bf67 | apt-devops-...	Off	10.0.2.0/24	–
public-0	subnet-0e43ff613607c7983	Available	vpc-029fdec5587e5bf67 | apt-devops-...	Off	10.0.0.0/24	–
private-1	subnet-074da233b5d5d5317	Available	vpc-029fdec5587e5bf67 | apt-devops-...	Off	10.0.3.0/24	–
–	subnet-0e562ce6f100fd96f	Available	vpc-0979ce9ab31158a9f	Off	172.31.16.0/20	–
–	subnet-0d064f884b14e9c46	Available	vpc-0979ce9ab31158a9f	Off	172.31.64.0/20	–

Route tables (4) Info

Last updated less than a minute ago | Actions

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
–	rtb-0992dae8740848688	2 subnets	–	No	vpc-029fdec5587e5bf67 | apt-d...
–	rtb-03ce18cce52a06608	6 subnets	–	Yes	vpc-0979ce9ab31158a9f
–	rtb-0c411f5ea858c45b4	2 subnets	–	No	vpc-029fdec5587e5bf67 | apt-d...
–	rtb-088dda67269e89cea	–	–	Yes	vpc-029fdec5587e5bf67 | apt-d...

Internet gateways (2) Info

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID
igw	igw-0495d0268759908ad	Attached	vpc-0979ce9ab31158a9f
apt-igw	igw-0b99145ac9d149d33	Attached	vpc-029fdec5587e5bf67 | apt-devops-vpc

Instances (4) Info

Last updated less than a minute ago | Connect | Instance state | Actions | Launch instances

Find Instance by attribute or tag (case-sensitive) | All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
apt-private-ec2	i-0abd18cd81b47d7e9	Running	t2.micro	Initializing	View alarms +	us-east-1b	–
apt-private-ec2	i-0802e67b68003babe	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	–
apt-private-ec2	i-08fe932fe4db9c897	Running	t2.micro	Initializing	View alarms +	us-east-1b	–
apt-private-ec2	i-03695f4acc7391ace	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	–

apt-alb

Actions

Details

Load balancer type: Application
Status: Active
VPC: vpc-029fdec5587e5bf67
Load balancer IP address type: IPv4
Scheme: Internet-facing
Hosted zone: Z35SXDOTRQ7X7K
Availability Zones: subnet-0d572efb03466425e us-east-1b (use1-az2); subnet-0e43ff613607c7983 us-east-1a (use1-az1)
Date created: December 5, 2025, 16:34 (UTC+05:30)
Load balancer ARN: arn:aws:elasticloadbalancing:us-east-1:474668386320:loadbalancer/app/apt-alb/0728dab21a06fb8d
DNS name Info: apt-alb-1340208238.us-east-1.elb.amazonaws.com (A Record)

Listeners and rules | Network mapping | Resource map | Security | Monitoring | Integrations | Attributes | Capacity | Tags

Listeners and rules (1) Info

Manage rules | Manage listener | Add listener

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS	Trust store
HTTP:80	• Forward to target group apt-tg : 1 (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable	Not applic

Elastic IP

Assigning Static Public IP

AWS does not allocate default EIP automatically.
Allocate an Elastic IP → associate with EC2 for static access.

Destroy / Cleanup

Deleting Resources

Run terraform destroy -auto-approve locally
Or trigger workflow with action: destroy

Optional: Manual deletion steps in AWS if needed (ASG → ALB → EC2 → VPC).